Process or Materials Review Request Form

Request Title: returning Si/SOI (Si-on-insulator) to "Clean Group" for CMOS FEOL after UV ozone photoresist descum in the Samco Ozone Cleaner at SNSF

Requestor: Jie Yao **Date: 1st May 2021**

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Company name/PI: Wavefront Vision Inc.

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** To return 4-inch and 6-inch Si or SOI (Si-on-insulator) to "Clean Group" for CMOS FEOL after UV ozone photoresist descum in the Samco Ozone Cleaner at SNSF (https://snsf.stanford.edu/facilities/fab/npc/general https://drive.google.com/file/d/1wvAVY9_HSq2EAyBl4MQeF-iBc4UgNcYC/view).
2. **Vendor/manufacturer info:** Samco, SNSF.
3. **Reason for request:** In order to process 6-inch Si and SOI wafers and to completely eliminate (instead of minimizing) plasma damage during oxygen descum of photoresist, the Samco UV Ozone Cleaner may be the best choice. We therefore request to allow the 4-inch and 6-inch Si/SOI wafers after photoresist descum in the Samco Ozone Cleaner at SNSF to be returned to the "Clean Group" for CMOS FEOL, including but not limited to Thermco furnaces, Tylan furnaces, AMAT Centurion, RCA cleaning baths, resist-piranha cleaning bath, resist-HF etch bath, resist-BOE etch bath, hot-phosphoric-acid etch, silicon rapid thermal annealing (RTA), Tencor p2 stylus profilometer, and other "Clean Group" instruments.

   Photoresist descum is a common, customary and natural step following photoresist development in the "All Group". Oxygen plasma is available in, for example, the Gasonics Aura Asher (the "Clean and Semi-clean Groups" and 4-inch wafers only), the Samco PC300 Plasma Etcher (the "Flexible Group"), or the Matrix Plasma Resist Strip (the "Flexible Group"), where the plasma damage can be minimized but never eliminated. 6-inch wafer size is another requirement exceeding the Gasonics' capabilities. We, therefore, propose the Samco UV Ozone Cleaner installed at SNSF for photoresist descumming on both 6-inch and 4-inch Si/SOI and to completely eliminate all plasma damage.
4. **Process Flow:** (1) We will aggressively physically wipe/scrub down the open-load chamber with water and isopropanol as much as possible and use UV ozone to etch back chamber immediately before UV ozone descum process. (2) We will descum our photoresist at 100 °C, lower than the photoresist post-exposure bake (PEB) and hard bake (HB) temperatures of 110 °C on the SVG developer track. (3) After UV ozone photoresist descum and its associated etch steps, we will decontaminate our Si/SOI wafers using the resist-piranha bath and the resist-HF50:1/BOE bath (combined similar but not identical to the IMEC-clean – the most advanced version of RCA cleaning) immediately before returning to any other instrument in the "Clean Group".
5. **Amount and form.** Not applicable.
6. **Storage:** Not applicable.
7. **Disposal**: Not applicable.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date:

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